Can AI Assistants Know What They Don't Know?

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Abstract

Recently, AI assistants based on large language models (LLMs) show surprising performance in many tasks, such as dialogue, solving math problems, writing code, and using tools. Although LLMs possess intensive world knowledge, they still make factual errors when facing some knowledge intensive tasks, like open-domain question answering. These untruthful responses from the AI assistant may cause significant risks in practical applications. We believe that an AI assistant's refusal to answer questions it does not know is a crucial method for reducing hallucinations and making the assistant truthful. Therefore, in this paper, we ask the question "**Can AI assistants know what they don't know and express them through natural language?**" To answer this question, we construct a model-specific "I don't know" (Idk) dataset for an assistant, which contains its known and unknown questions, based on existing open-domain question answering datasets. Then we align the assistant with its corresponding Idk dataset and observe whether it can refuse to answer its unknown questions after alignment. Experimental results show that after alignment with Idk datasets, the assistant can refuse to answer most its unknown questions. For questions they attempt to answer, the accuracy is significantly higher than before the alignment. [1]

1 Introduction

Large language models (Brown et al., 2020; Chowdhery et al., 2023; Zeng et al., 2023; Touvron et al., 2023) possess extensive world knowledge and demonstrate capabilities in numerous natural language tasks that smaller models lack (Wei et al., 2022b). Recently, there are many artifical intelligence chat assistants built on large language models, capable of helping users accomplish various tasks in the daily life, providing a satisfactory user experience (Ouyang et al., 2022; OpenAI, 2022; Anthropic, 2023; Sun et al., 2023; Baichuan, 2023; Qwen-Team, 2023). Although these chat assistants can communicate frequently with users, they are prone to hallucinations (Shuster et al., 2021; Zhang et al., 2023b; Cheng et al., 2023), such as including factual errors in their generated responses (Wang et al., 2023b) or imitating human falsehoods in training corpus (Lin et al., 2022a), some of which are difficult for users to detect. These untruthful responses could potentially harm society and also diminish the credibility of AI assistants.

An AI assistant aligned with human values should be truthful (Evans et al., 2021), which means that it needs to provide accurate information consistent with the real world. When the assistant's output contains factual errors, it indicates that it may lack the corresponding knowledge internally, yet it fails to express the unknowns and refuse to give a answer.

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[1]We will release our code, data and models at https://github.com/OpenMOSS/Say-I-Dont-Know.

	Unknowns	Knowns
Known	Known Unknowns: Things the AI knows it doesn't know.	Known Knowns: Things the AI knows it knows.
Unknown	Unknown Unknowns: Things the AI doesn't know it doesn't know.	Unknown Knowns: Things the AI doesn't know it knows.

Figure 1: Knowledge quadrants of an AI assistant. "Unknowns" represents what the AI does not actually know. "Knowns" represents what the AI actually knows. "Known" represents what the AI believes it knows. "Unknown" represents what the AI believes it does not know.

However, a truthful AI assistant should be aware of what it knows and what it does not know, and be able to communicate this to the user. For questions that are known, the AI assistant should provide users with accurate information. For questions that are unknown, the AI assistant should avoid giving answers. Therefore, in this paper, we explore the question "Can AI assistants know what they don't know and express them through natural language?".

The AI assistant's perception of its own knowledge can be represented through knowledge quadrants (Yin et al., 2023b). The knowledge quadrant is a partition which can divide the knowledge into four categories: Known Knowns, Known Unknowns, Unknown Knowns and Unknown Unknowns, as shown in Figure 1. Known Knowns is crucial for a truthful AI assistant, as it relies on its own knowledge to provide accurate and reliable responses. The more knowledge that falls under the category of Known Knowns, the more helpful the AI assistant becomes. We use Ik-Ik (I know I know) to represent Known Knowns. Besides, we argue that a truthful AI assistant should also be aware of and express its lacks in certain knowledge. Specifically, it should admit when it doesn't have information on a topic or when the information is not certain to maintain truthfulness. This part of knowledge falls under the category of Known Unknowns. We use Ik-Idk (I know I don't know) to represent Known Unknows. Unknown Unknowns and Unknown Knowns will cause untruthful and helpless generations. We use Idk-Idk (I don't know I don't know) and Idk-Ik (I don't know I know) to represent Unknown Unknowns and Unknown Knowns respectively. To make AI assistants truthful, we need to teach AI assistants to know what they know and what they do not know to convert Unknown Knowns and Unknown Unknowns to Known Knowns and Known Unknowns.

Our approach is to align an AI assistant (like llama-2-7b-chat) with a model-specific **"I don't know" (Idk)** dataset which contains the assistant's known and unknown questions. We construct the Idk dataset based on an existing knowledge-intensive open-domain question answering dataset, TriviaQA (Joshi et al., 2017). We determine whether an assistant knows the answer to a question by evaluating its average accuracy across multiple responses to that question. Questions that the assistant answers incorrectly multiple times are marked as ones it does not know, and a template for refusal to answer is annotated. For questions that the assistant answers correctly multiple times, a correct answers it generates are used for the annotated answer. The assistant's accuracy threshold for being considered knowing the answer to a question is a hyperparameter, which we call **Ik threshold**. We discuss the details of constructing the Idk dataset in Section 3.1.

In order to teach AI assistants to know what they don't know, we conduct extensive experiments to exploit the most effective method, including prompting, supervised fine-tuning and preference-aware optimization. For prompting, we instruct the assistant to

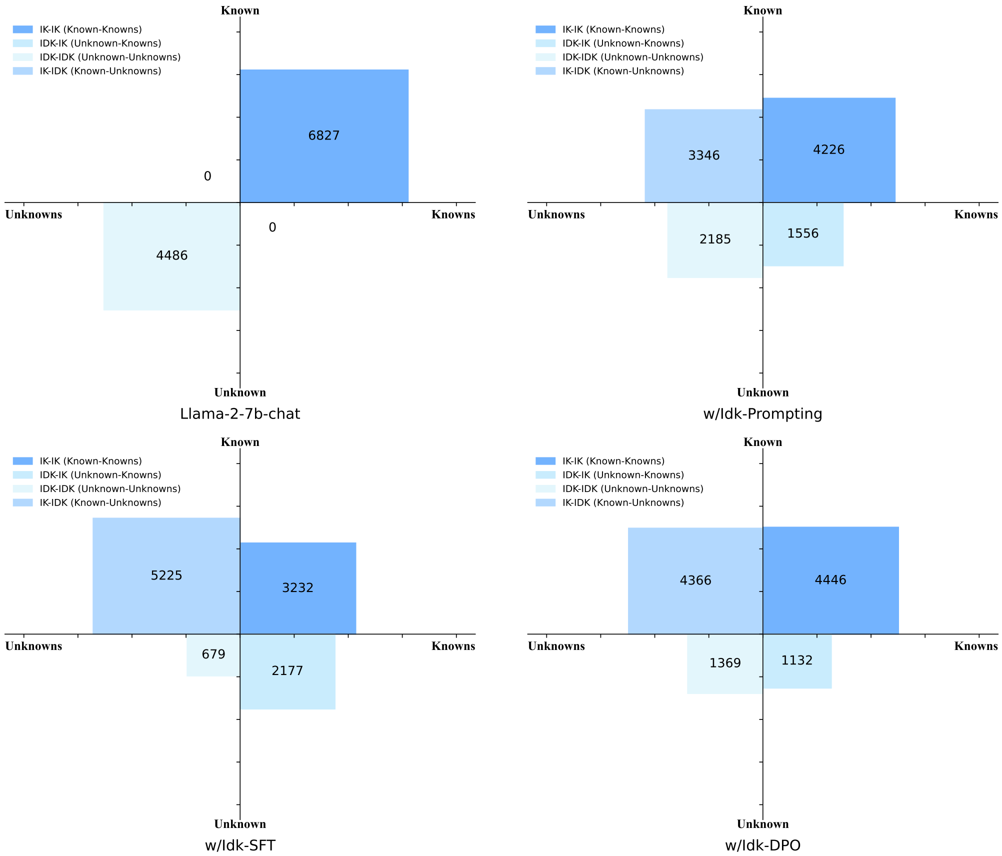

Figure 2: Knowledge quadrants of AI assistants on the Idk dataset (Ik threshold=1.0). **IK-IK** represents the AI answers the questions correctly. **IDK-IK** represents the AI knows the answer but refuses to respond to the question. **IDK-IDK** represents the AI answers the question incorrectly. **IK-IDK** represents the AI doesn't know the answer and refuses to respond to the question. **w/Idk-Prompting:** Using prompting can transform certain IDK-IDK questions to IK-IDK questions. **w/Idk-SFT:** Idk-SFT allows the model to refuse to answer more questions it does not know, but it also tends to make the model more convervative, leading to incorrect refusals to answer some questions that it actually knows. **w/Idk-DPO:** Using preference-aware optimization, like DPO, can alleviate the model's excessive conservatism and reduce the number of IDK-IK questions.

refuse answering questions it does not know through a prompt. For supervised fine-tuning (SFT), we directly fine-tune the original assistant using our Idk datasets. For preference-aware optimization, we use best-of-n sampling (BoN), proximal policy optimization (PPO) (Schulman et al., 2017; Ouyang et al., 2022), direct preference optimization (DPO) (Rafailov et al., 2023) and hindsight instruction relabeling (HIR) (Zhang et al., 2023a). We demonstrate some representative results in Figure 2.

The original model (llama-2-7b-chat) can be considered as lacking the ability to recognize questions it does not know[2]. It may guess an answer even it lacks the knowledge. As shown, there are many IDK-IDK questions, making the assistant untruthful. Instructing the model to refuse answering unknown questions through a prompt can be effective to some extent, but there are still numerous IDK-IK and IDK-IDK questions. After supervised fine-tuning

[2]We conducted a search for keywords such as "I don't know", "not sure", "Sorry" in the responses of Llama-2-7b-chat and found that only a very small number of responses contained these keywords.